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Senior Python Full Stack Developer

PROFESSIONAL SUMMARY:

- ◆ **Senior Full Stack Python Developer** with over 10 years of experience in designing, developing, and optimizing enterprise-level applications across various domains, including finance, healthcare, and e-commerce.
- ◆ Proficient in Python Development with a strong background in building RESTful web services using Flask, Django, and FastAPI, delivering scalable and efficient backend solutions.
- ◆ Experienced in Frontend Development using modern frameworks like ReactJS and Angular, creating dynamic and responsive UIs with a focus on performance and user experience.
- ◆ Skilled in Microservices Architecture, developing and deploying scalable microservices using Python, Flask, and Spring Boot, and integrating them via RESTful APIs and AWS services.
- ◆ AWS Cloud Expertise in deploying and managing applications using AWS EC2, S3, RDS, Lambda, Route 53, and Kubernetes, ensuring high availability, scalability, and disaster recovery.
- ◆ Extensive Experience with CI/CD Pipelines, automating build, testing, and deployment processes using Jenkins, Docker, Terraform, and Maven for seamless and efficient development cycles.
- ◆ Strong Database Management Skills in both SQL (PostgreSQL, MySQL) and NoSQL (Cassandra) databases, with expertise in SQLAlchemy ORM, complex query optimization, and schema design.
- ◆ Proficient in Asynchronous Programming and multithreading in Python, using asyncio, threading, and concurrent.futures to develop high-performance, responsive applications.
- ◆ In-depth Knowledge of JavaScript and AJAX, implementing client-side validations, asynchronous data handling, and SPA development using React Router and Angular services.
- ◆ Versatile in Object-Oriented and Functional Programming, applying best practices in Python for clean, maintainable, and efficient code using libraries like functools and itertools.
- ◆ Experienced in Version Control and Collaboration, utilizing Git and GitHub/Bitbucket for source code management and collaborative development within cross-functional teams.
- ◆ Proficient in Automated Testing and Quality Assurance, using PyTest, Unittest, and JMeter to ensure robust and reliable code, with a strong focus on continuous validation and quality delivery.
- ◆ Strong Analytical and Problem-Solving Skills, with a proven track record of identifying system inefficiencies, optimizing application performance, and enhancing user experience.
- ◆ Effective Communicator and Team Player, with experience leading projects, mentoring junior developers, and collaborating with stakeholders to ensure alignment with business goals.
- ◆ Dedicated to Continuous Learning, staying updated with the latest industry trends, tools, and technologies, and applying them to drive innovation and improvement in development practices.

Technical Skills:

Category	Technologies/Tools
Programming Languages	Python, JavaScript, HTML, CSS, SQL, PL/SQL, Golang, Java
Python Frameworks	Flask, Django, FastAPI, Flask-RESTful, Flask-Security, SQLAlchemy, asyncio, threading, concurrent.futures, functools, itertools
Java Frameworks	Spring Boot, Spring MVC, Spring Ribbon
Frontend Frameworks/Libraries	ReactJS, Angular (4, 6, 15), jQuery, Bootstrap, AJAX, JSX, JavaScript API, React Native, AngularJS, React Router
Web Services	RESTful APIs, SOAP, Flask/Django Routes, Apache Tomcat, Spring Boot
Cloud Platforms & Services	AWS (EC2, S3, RDS, Lambda, Route 53, ELB, Auto Scaling, SNS), Kubernetes, OpenShift, AWS Lambda, Pivotal Cloud Foundry (PCF)
Containers & CI/CD	Docker, Docker Compose, Jenkins, Maven, Terraform, Pipenv, Git, GitHub, Bitbucket, Jenkins, Apache Maven

Database Management	PostgreSQL, MySQL, Cassandra, SQLAlchemy ORM, SQL Joins, Stored Procedures, Query Optimization, Schema Design
Asynchronous Programming	Python async/await, multithreading, multiprocessing, asyncio, concurrent.futures
Version Control	Git, GitHub, Bitbucket
Testing Frameworks	PyTest, Unittest, JMeter, Nose, SoapUI
Message Queues & Streaming	Kafka, RabbitMQ, Zeep, Requests
Microservices	Flask, Django, FastAPI, Python, Spring Boot, Kafka, AWS API Gateway
Object-Oriented & Functional Programming	Python OOP, Java OOP, Lambda functions, functools, itertools
Security & Authentication	Flask-Security, OAuth, Exception Handling
Web Servers	Apache Tomcat, WSGI, Apache, Nginx
DevOps & Infrastructure	Terraform, Docker, Kubernetes, Jenkins, Terraform, Route 53, SNS, Pivotal Cloud Foundry, AWS Infrastructure Management
Design & Documentation	UML (Use Cases, Sequence diagrams, Class diagrams, Page flow diagrams), Lucidchart, Visual Paradigm
Automation & Scripting	Python Automation Scripts, Shell Scripting, UNIX System Administration, Machine Learning
Methodologies	Agile, SCRUM

PROFESSIONAL EXPERIENCE:

Client: American Express. Phoenix, AZ

September 2022 – Present

Role: Sr. Python Developer

Responsibilities:

- Developed and maintained RESTful web services using Python and Flask/Django to deliver JSON responses for front-end integration.
- Upgraded and maintained JavaScript libraries and widgets to ensure consistent data management across forms, regardless of development time.
- Created interactive UIs with ReactJS, leveraging one-way data flow, Virtual DOM, JSX, and React Native concepts.
- Developed and deployed machine learning models using **scikit-learn** and **TensorFlow** to detect fraudulent transactions in real-time.
- Designed automated machine learning pipelines using **AWS SageMaker** for model training, hyperparameter tuning, and deployment.
- Built and managed ReactJS components, including Forms, Events, Keys, Router, Animations, and implemented the Flux architecture.
- Utilized React Router to develop Single Page Applications (SPAs), ensuring smooth navigation and dynamic content rendering.
- Developed and customized modules and widgets using React JS, CSS, and HTML, ensuring cross-browser compatibility.
- Implemented Flask/Django routes to handle HTTP requests, connecting with various services and databases for data processing.
- Configured Flask/Django applications with RESTful endpoints, implementing security features and integrating with external APIs.
- Utilized SQLAlchemy for ORM, performing database operations using Python with complex queries, migrations, and schema management.
- Developed REST APIs to process data between databases and external services, ensuring seamless data flow across platforms.
- Created Microservices using Flask/Django for efficient data processing and integration with other microservices via RESTful APIs.
- Worked with Docker to containerize Python applications, streamlining deployment processes across various environments.
- Developed and managed Kafka consumers and producers in Python, processing messages and transforming data formats for storage in SQL and NoSQL databases.
- Integrated SOAP and RESTful web services, Python using libraries like Zeep and Requests, facilitating communication with external services.
- Deployed applications on AWS using services like EC2, S3, and RDS, with additional configuration and management via Route 53 and ELB.

- Configured and maintained Apache Tomcat for deploying Java-based applications and integrated Python-based services via WSGI.
- Implemented and monitored Cassandra clusters, focusing on data modeling, performance tuning, and managing large datasets in a distributed environment.
- Managed project dependencies and deployments using tools like Pipenv, Docker Compose, and automated CI/CD pipelines with Jenkins.
- Utilized Git for version control, maintaining code integrity and collaborating with teams using GitHub for source code management.
- Conducted unit testing using PyTest, ensuring code quality and reliability, and implemented logging with Python's logging module for monitoring and debugging.

Environment: Python, Flask, Django, ReactJS, JavaScript, HTML, CSS, RESTful, JSON, REST APIs, Microservices, React Router, JSX, Virtual DOM, ORM, SQLAlchemy, Docker, Compose, Kafka, API Integration, SOAP (Zeep), AWS (EC2, S3, RDS, Route 53, ELB), Apache Tomcat, Cassandra, SQL, NoSQL, Git, GitHub, CI/CD, Jenkins, Pipenv, PyTest, Python logging, Automated pipelines

Client: CVS Health. Woonsocket, RI

Jan 2019 – August 2022

Role: Sr. Python Developer

Responsibilities:

- Developed a Medical Imaging System know as PACS (Picture Archiving and Communication System), a medical imaging technology used in healthcare organizations to securely store, retrieve, manage, distribute, and present medical images.
- Participated in documenting (analysis, specification, design), implementing, and testing phases of the Software Development Life Cycle (SDLC).
- Utilized Python, Flask, Django, and RESTful APIs for backend development.
- Leveraged Angular 15 for the frontend, optimizing application performance and maintainability.
- Developed and maintained microservices using FastAPI, optimizing for fast startup times and low memory footprint.
- Applied functional programming principles in Python, using libraries like functools and itertools to write clean, concise, and efficient code.
- Ensured compliance with DICOM standards for handling, storing, printing, and transmitting medical imaging information, ensuring interoperability between different imaging devices and PACS.
- Designed and developed scalable and fault-tolerant microservices using Python and Flask, ensuring high availability and reliability.
- Leveraged AI tools within the system to process image, text, and numeric data, delivering enhanced diagnostic results.
- Extensive experience with testing frameworks and tools for Python, such as PyTest, Unittest, and Nose, ensuring robust and reliable code.
- Implemented Flask-Security for API-level authentication and authorization, reducing development time while enhancing security.
- Developed and customized solutions using Python for transaction services clients, enhancing compliance and risk management capabilities.
- Worked on process modelling using BPMN tools integrated with Python applications, defining tasks, events, and gateways to create robust process flows.
- Built and deployed services on AWS, including importing necessary volumes, launching EC2 instances, creating security groups, and configuring auto-scaling, load balancers, Route 53, and SNS.
- Involved in designing and implementing disaster recovery plans using EC2 and related AWS services, ensuring business continuity.
- Implemented Python's async/await syntax to manage asynchronous tasks, resulting in more efficient and responsive applications.
- Configured and deployed Python applications on Kubernetes and OpenShift, ensuring scalable and resilient cloud-native solutions.
- Implemented business rule engines within the system for transaction monitoring and fraud detection, enhancing security.
- Utilized Python's exception handling features to increase application reliability and reduce runtime errors.
- Created multiple Kubernetes clusters on AWS and deployed Docker images on top of these clusters for testing purposes.
- Involved in map development activity using Python-based AI tools.
- Followed best practices when using Python's lambda functions, ensuring readability, avoiding overuse, and maintaining clarity in code.
- Used Python Kafka libraries to process messages efficiently on Kafka Cluster setup.
- Employed various performance optimization techniques in Python web applications, such as profiling, caching, and load balancing.

- Developed cross-platform server-side applications using Python, Django, and ReactJS to achieve scalability in real-time web applications.
- Used Apache Maven for build automation and Jenkins for continuous integration, ensuring smooth and efficient development cycles.
- Developed the application using the SCRUM methodology, participating in daily scrum meetings to discuss progress and challenges.
- Leveraged Python's multiprocessing and concurrent.futures for parallel processing and multi-threaded operations, ensuring optimal performance in compute-intensive tasks.
- Collaborated with medical professionals to discuss imaging results and recommend further actions or treatments, ensuring patient care.
- Ensured compliance with safety protocols and maintained patient confidentiality throughout the imaging and diagnostic process.
- Provided guidance and mentorship to junior radiologists and medical staff, fostering a collaborative and educational environment.
- Conducted a variety of imaging procedures, interpreting results to assist in patient diagnosis and improve diagnostic accuracy.

Environment: PACS, DICOM standards, SCRUM, Agile, Python, JavaScript, Flask, Django, FastAPI, Angular 15, ReactJS, Microservices, FastAPI, Flask, Python, AI tools for image, text, and numeric data, PyTest, Unittest, Nose, Flask-Security, API-level auth, Python's async/await, profiling, AWS (EC2, auto-scaling, load balancers, Route 53, SNS), Kubernetes, OpenShift, Kubernetes, Docker, Terraform, AWS, BPMN, Python's exception, Python Kafka libraries, Apache Maven, Jenkins, Kafka Cluster, PostgreSQL

Client: Fannie Mae, Washington, District of Columbia

July 2017 – December 2018

Role: Sr. Python Full Stack Developer

Responsibilities:

- System Optimization by Analyzed the complete system flow and identified various gaps in the current system.
- Designed a new, streamlined flow for faster report creation and posting, replacing complex logic with simpler, more stable code structures.
- Captured business requirements, contributed to design, development, and testing phases of the application, and ensured alignment with user needs for end to end development
- Prepared user stories corresponding to features taken from the backlog, driving the iterative development process.
- Applied both procedural and object-oriented programming approaches in Python to develop robust applications.
- Provided technical support throughout all phases of testing and quality assurance, ensuring smooth implementation and delivery.
- Proactively address bug fixing and maintenance tasks, ensuring the product's stability and reliability post-deployment.
- Utilized Go (Golang) to log different host system events and alert information to a Cassandra database, ensuring comprehensive monitoring.
- Built interactive and responsive web pages using AJAX, jQuery, AngularJS, and JavaScript.
- Enhanced user experience with Angular 4's Http services for asynchronous validations and client-side functionalities.
- Worked extensively with Angular 6 to build a cross-device/browser web application for location sharing.
- Worked on creating log-in authentication modules using AngularJS and JavaScript.
- Developed a cross-device/browser web application for location sharing utilizing Angular 6, JavaScript API.
- Worked on Java 8 features like Lambda expressions and Stream API for Bulk data operations on Collections which would increase the performance of the Application.
 - Used two-way data binding, event handling, and AJAX calls to deliver a seamless user experience.
 - Managed AWS infrastructure using Terraform, automating the deployment and scaling of instances, and maintaining infrastructure consistency.
 - Developed backend services using Python and Node.js, integrated with third-party services such as Firebase, Kinvey, and Parse.
 - Hosted and consumed RESTful web services, ensuring scalable cloud-based solutions.
 - Designed and implemented microservices using Python and Spring Boot, coupled via RESTful APIs and AWS API Gateway for a clear separation of functionalities.
 - Provisioned and managed different application environments on AWS using Docker, Bash, and Terraform.
 - Deployed Docker-contained services onto AWS EC2 instances and utilized AWS Lambda for serverless computing.
 - Executed comprehensive load testing scenarios using JMeter, addressing stress, scalability, failover, and load performance to ensure application robustness.

- Used Git source control (Bitbucket) to track and manage development progress, ensuring seamless collaboration and code integration across the team.
- Designed Kafka clusters, implemented producer and consumer applications, and managed PostgreSQL databases, including backup, recovery, and performance optimization.
- Deployed, scaled, and configured microservices in Pivotal Cloud Foundry (PCF), writing manifest files for efficient management.
- Employed Spring Ribbon for load balancing, implemented Circuit Breaker patterns with Hystrix for fail-safe methodology, and ensured high availability of microservice applications.
- Implemented OAuth for secure access to cloud storage services like Google Drive and Dropbox, enhancing data security.
- Automated build and deployment processes using Jenkins and Maven, enabling continuous integration and smooth production releases.

Environment: System flow analysis, Simplified code structures, Agile, User stories, End-to-end development, Python, JavaScript, Go (Golang), Java, Angular 4, Angular 6, Flask, Spring Boot, AJAX, jQuery, AngularJS, JavaScript, HTML, CSS, Python, Node.js, Spring Boot, Flask, Java 8 (Lambda expressions, Stream API), Cassandra, PostgreSQL, Firebase, Kinvey, Parse, AWS (EC2, Lambda, API Gateway), Docker, Terraform, Pivotal Cloud Foundry (PCF), Terraform, Bash, Docker, YAML, JMeter, Python, Spring Boot, RESTful APIs, Spring Ribbon, Hystrix, Jenkins, Maven, Git (Bitbucket), Automated, Kafka, Kafka producer, OAuth, Git (Bitbucket), RESTful web services, JSON, XML

Client: Kronos, TX
2017

March 2015 – July

Role: Java/J2EE Developer

Responsibilities:

- Worked on analysis, design, and development phases of the Software Development Life Cycle (SDLC), ensuring alignment with project goals and requirements.
- Utilized Agile Methodology for managing the end-to-end project lifecycle, ensuring iterative development and timely delivery of features.
- Developed UI modules using HTML, Flask, jQuery, Bootstrap, JavaScript, and CSS, creating responsive and user-friendly interfaces.
- Implemented AJAX in JSON format for asynchronous request handling, enabling partial page refreshing and enhancing user experience.
- Developed reusable and interoperable web service modules based on SOA architecture using RESTful APIs, ensuring efficient communication between services.
- Extensively worked with Flask, HTML, and XML to design the GUI framework and presentation layer, with client-side validations handled using JavaScript.
- Created configuration files in YAML and JSON, managing application settings and dependencies.
- Transitioned from Spring MVC to Flask for lightweight and efficient web application development, implementing Flask blueprints for modular design.
- Used Flask extensions for dependency injection and managed cross-cutting concerns like logging, security, and transaction management.
- Worked with SQL Alchemy as an Object-Relational Mapping (ORM) tool for backend data persistence, ensuring efficient data interaction.
- Developed application database objects, including stored procedures, using SQL and PL/SQL to interact with a PostgreSQL database.
- Worked with both MySQL and PostgreSQL databases to save and retrieve data related to payment information and user subscriptions.
- Wrote optimized SQL queries and stored procedures, enhancing database performance by analyzing and improving query execution plans.
- Designed and implemented RESTful web services using Flask-RESTful, ensuring efficient communication between client and server.
- Worked with Jenkins for Continuous Integration (CI) builds, automating testing and deployment processes.
- Configured Git repositories to integrate with Jenkins, triggering builds automatically upon code commits and ensuring smooth CI/CD workflows.
- Managed server configurations and message queue setups, using tools like RabbitMQ for reliable message queuing.
- Used JSON for data exchange in RESTful services and XML for configurations and data storage, employing Python libraries to parse and manipulate these formats.
- Utilized Python's logging module to capture runtime exceptions and debug issues effectively, ensuring smooth application performance.
- Developed unit test cases using PyTest, ensuring code reliability and robustness. Integrated testing into the CI pipeline for continuous validation.

Environment: SDLC, Agile Methodology, Python, JavaScript, SQL, PL/SQL, Flask, Spring MVC, HTML, jQuery, Bootstrap, JavaScript, CSS, AJAX (JSON format), Flask, Flask-RESTful, SQLAlchemy (ORM), Flask Blueprints, PostgreSQL, MySQL, RESTful APIs, SOA Architecture, Flask-RESTful, YAML, JSON, XML, Jenkins, Git, CI/CD workflows, RabbitMQ, Python's logging module, PyTest, Git, Jenkins

Client: Fresh Direct, NY

May 2013 – Feb 2015

Role: Software Engineer

Responsibilities:

- Conducted analysis and design for the project, utilizing MVC (Model-View-Controller) architecture and design patterns to structure the application effectively.
- Optimized Python code and database queries to maximize the performance of the enterprise application, ensuring efficient data processing and retrieval.
- Implemented client-side validations using JavaScript and integrated with Python-based web frameworks like Flask or Django for seamless user interactions.
- Developed and implemented asynchronous, multi-threaded file processing using Python's threading and multiprocessing modules, improving application responsiveness and efficiency.
- Designed and implemented a common UI layout for headers, footers, and menus using Flask templates and Bootstrap for consistent and responsive user interfaces.
- Tested RESTful web services using tools like Postman and SoapUI, validating request and response data in JSON and XML formats.
- Created UML Use Cases, Sequence diagrams, Class diagrams, and Page flow diagrams using tools like Lucidchart or Visual Paradigm to document and communicate the system design.
- Designed and developed user interfaces using HTML, Flask, and Bootstrap, allowing users to interact with dynamic content and application features.
- Developed automation scripts using Python for builds, deployments, and other repetitive tasks, streamlining the development process.
- Developed backend logic using Python, Flask, and SQLAlchemy for interacting with a PostgreSQL or MySQL database, enabling dynamic data retrieval and processing.
- Generated and displayed day-wise transaction reports in the browser using Flask and Jinja2 templates, providing users with detailed and organized data.
- Wrote complex SQL join queries and stored procedures for reporting, handling various reporting requirements with efficiency.
- Authored detailed design documents with UML specifications, outlining system architecture and component interactions.
- Packaged and deployed Python-based applications into production environments, such as Apache or Nginx, ensuring reliable operation.
- Performed proactive UNIX system administration tasks, including patch management, software upgrades, and shell scripting for automation.
- Used Python's file I/O operations to create and automate reports, which were then emailed to managers at regular intervals, ensuring timely updates.

Environment: Python, JavaScript, Flask, Django, MVC (Model-View-Controller), Design Patterns, HTML, Bootstrap, JavaScript, Python Threading, Python Multiprocessing, Flask, SQLAlchemy, PostgreSQL, MySQL, Postman, SoapUI, UML, Lucidchart, Visual Paradigm, Jinja2 (for Flask), Python Scripting, UNIX, Shell Scripting, Apache, Nginx, SQL, Python File I/O